

LTSpice Notch Quality Report

CROSS COUPLED2

Auto-detected swept parameters and quality-factor ($Q = f_0 / BW$ @ passband-3 dB) analysis

Run summary

Signal columns analyzed: 1

Total simulation steps : 72

V(act+,act-)
swept parameters: Cp, Itail

Report contents (per signal column):

- 1. All curves — Bode plot (magnitude + phase)
- 2. Magnitude grouped by each swept parameter (one subplot per fixed value)
- 3. Phase grouped by each swept parameter (one subplot per fixed value)
- 4. Quality-factor table for that column

End of report (extra analyses):

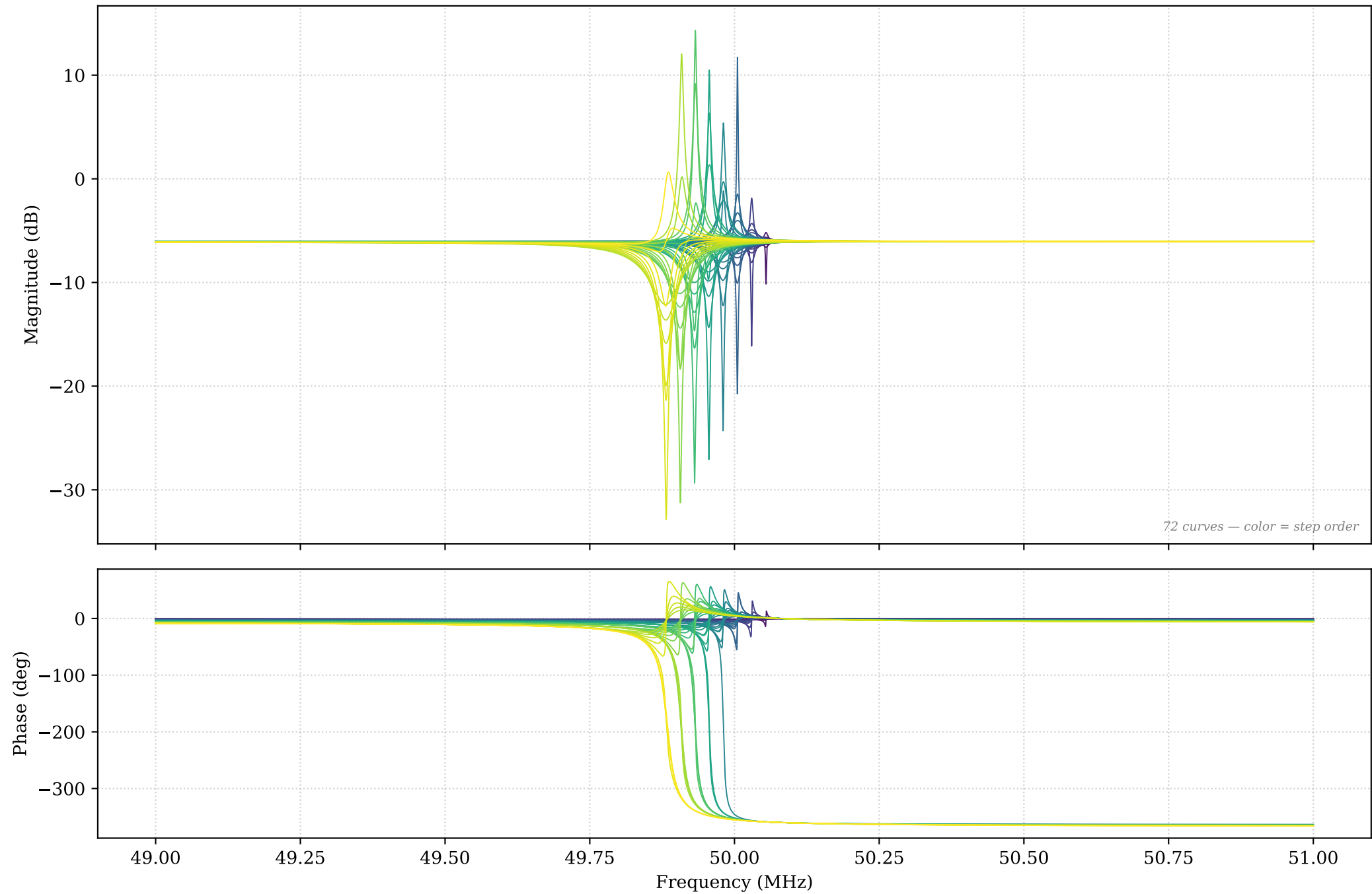
- A. Cross-signal comparison (act vs pas, if both present)
- B. Q vs each swept parameter (one line per other-param combo)
- C. BW vs each swept parameter
- D. Q heatmap for V(act): rows $\times$ columns swept params, colored
- E. Combined quality-factor table across all signals

Signal: $V(\text{act+}, \text{act-})$

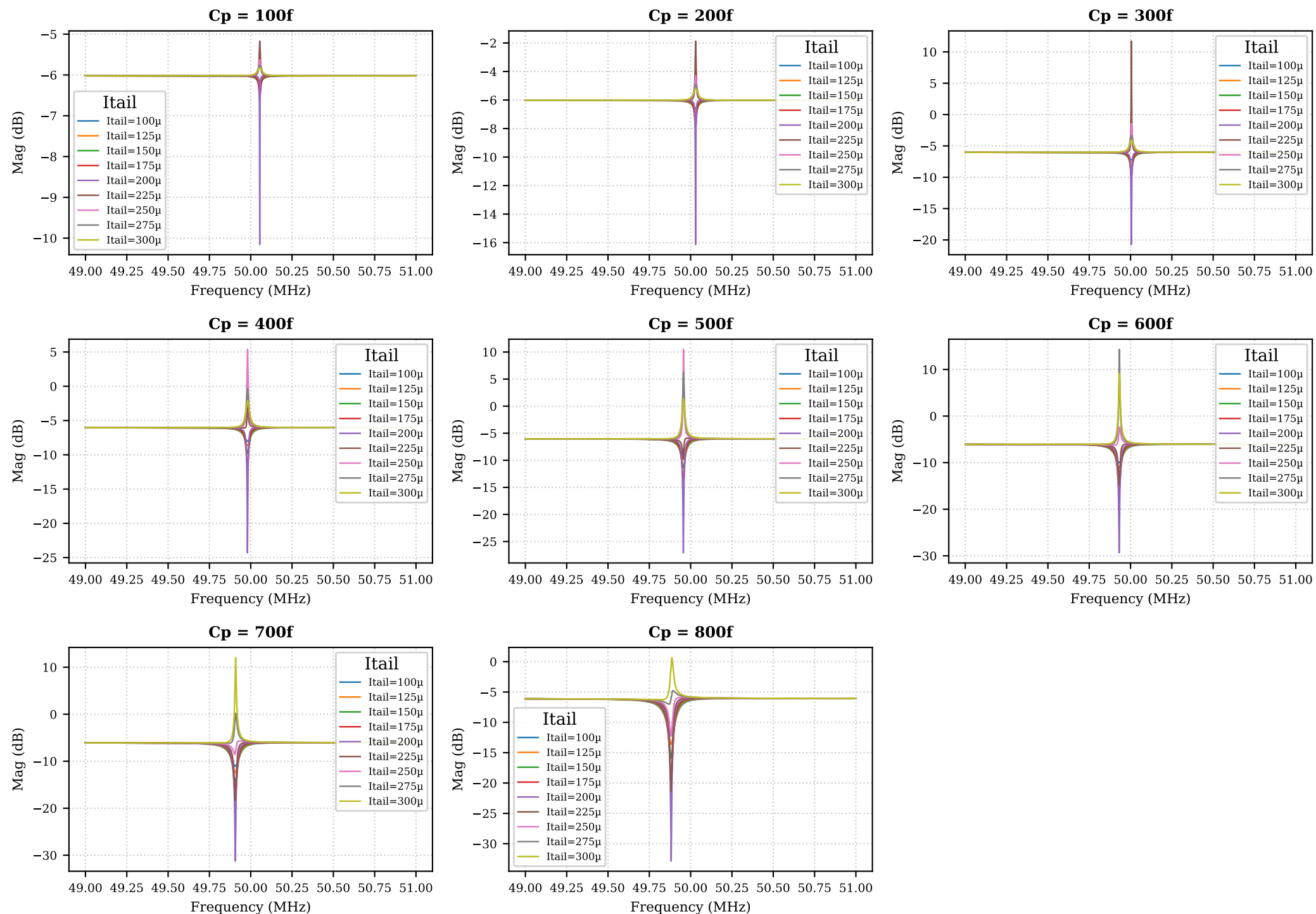
Swept parameters: C_p , I_{tail}

Steps: 72

V(act+,act-) — All curves (72 steps) — Bode

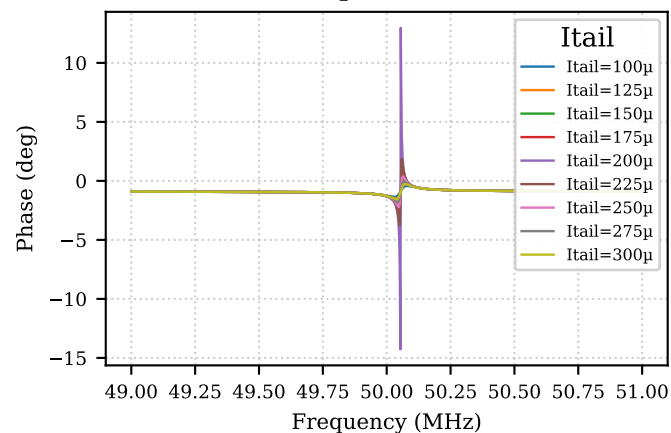


V(act+,act-) — Magnitude grouped by Cp (each subplot fixes Cp; legend = Itail)

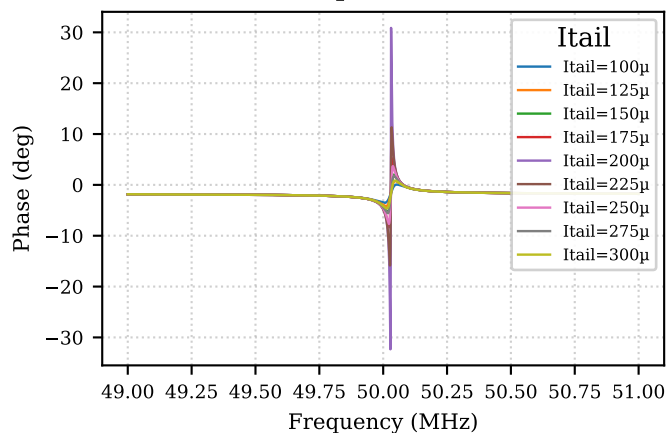


V(act+,act-) — Phase grouped by Cp (each subplot fixes Cp; legend = Itail)

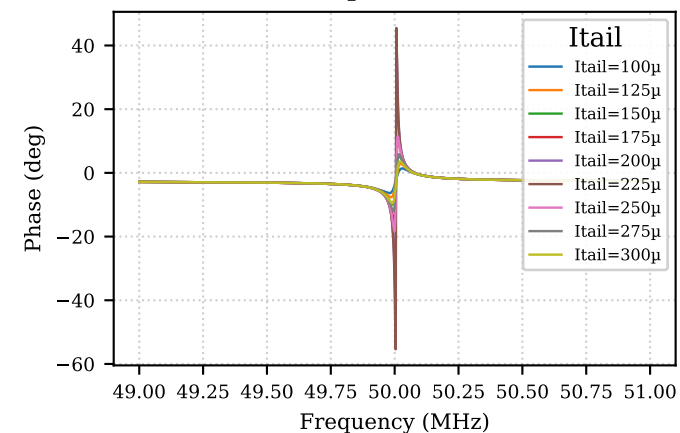
Cp = 100f



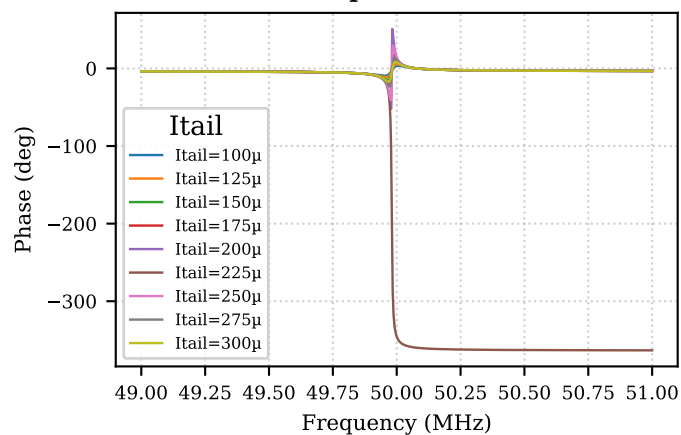
Cp = 200f



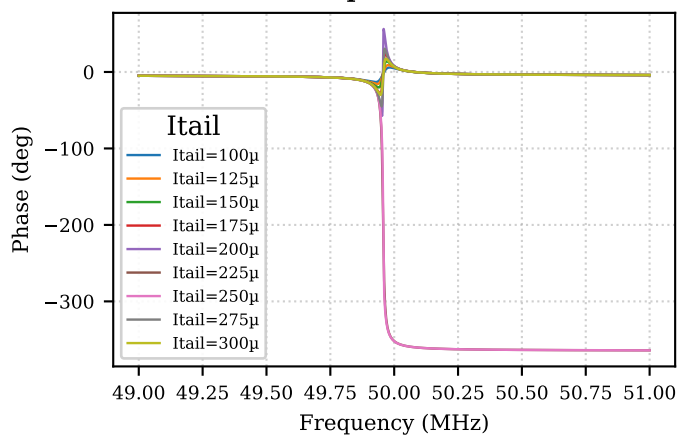
Cp = 300f



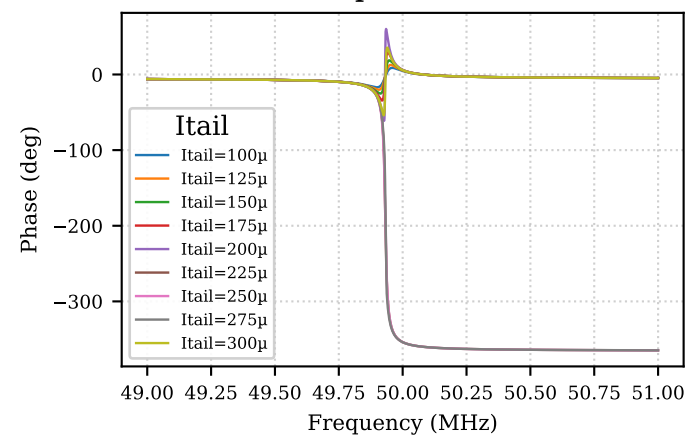
Cp = 400f



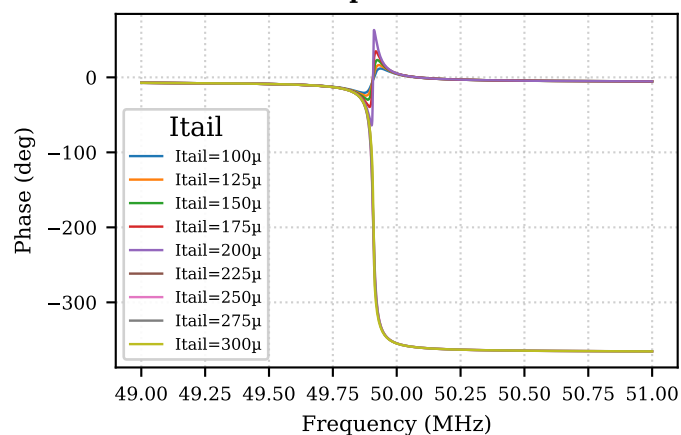
Cp = 500f



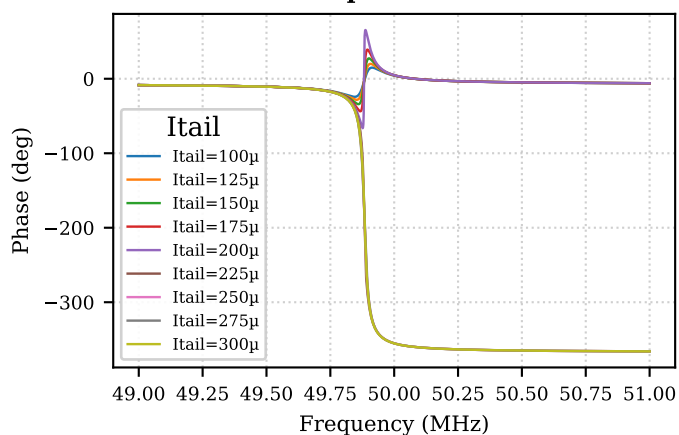
Cp = 600f



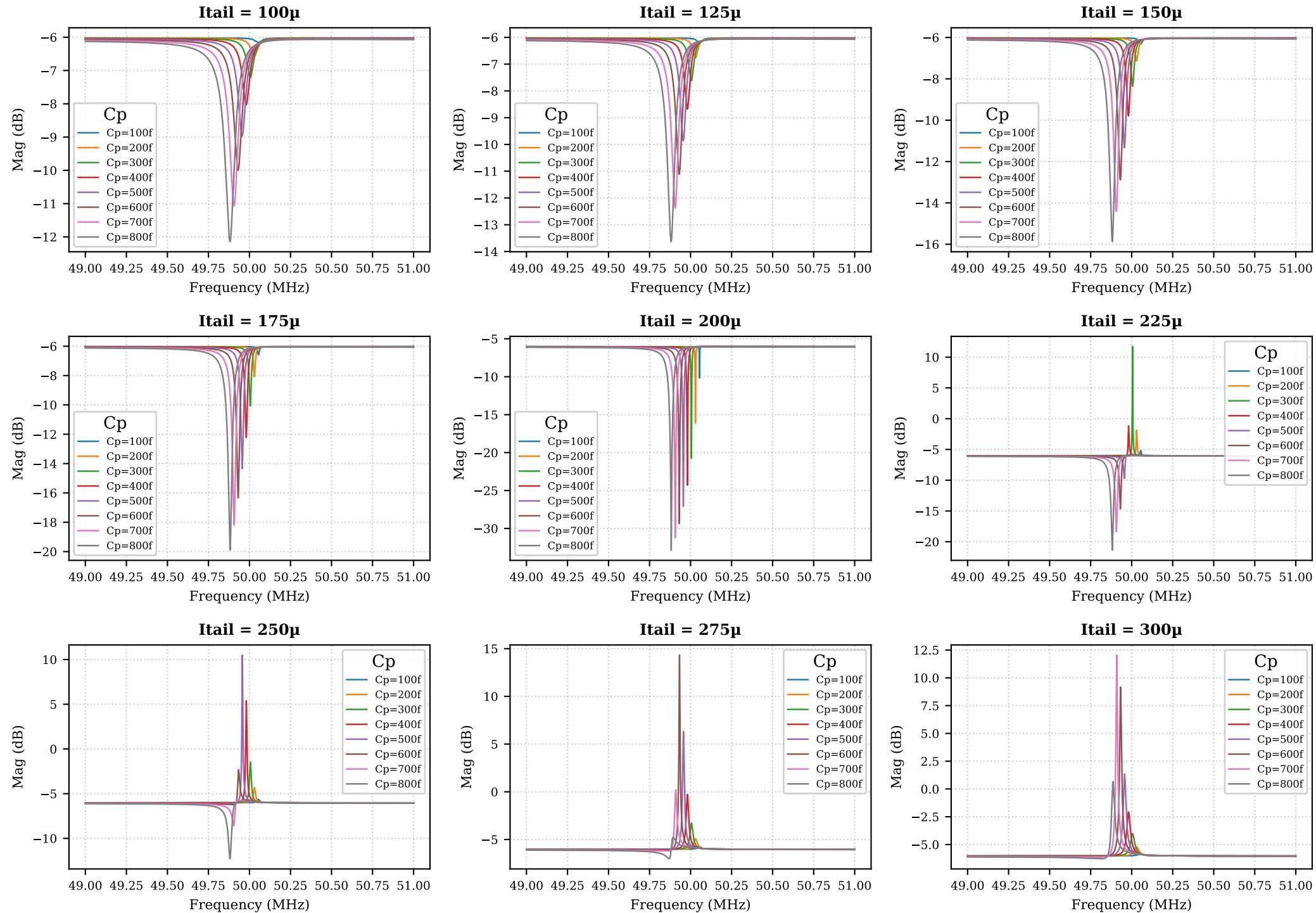
Cp = 700f



Cp = 800f

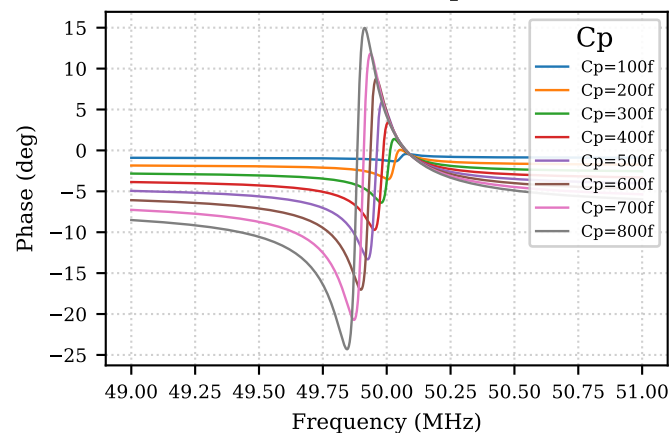


$V(\text{act+}, \text{act-})$ — Magnitude grouped by I_{tail} (each subplot fixes I_{tail} ; legend = C_p)

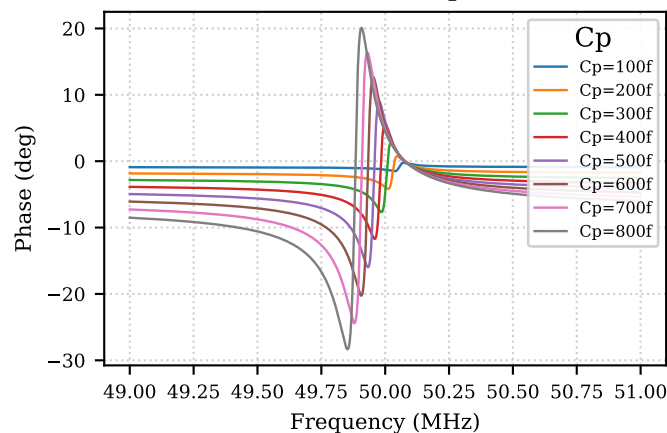


V(act+,act-) — Phase grouped by Itail (each subplot fixes Itail; legend = Cp)

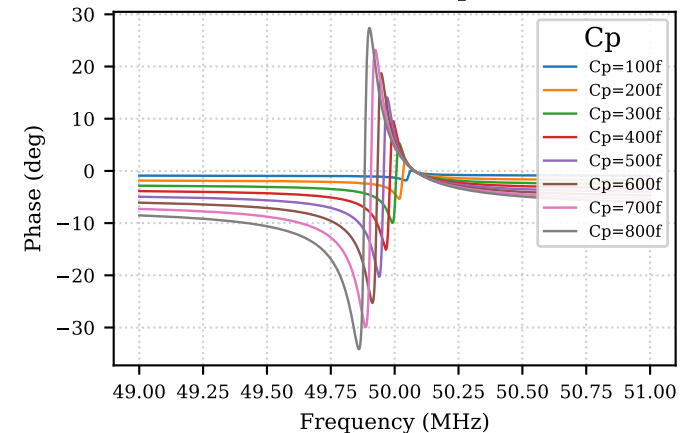
Itail = 100 μ



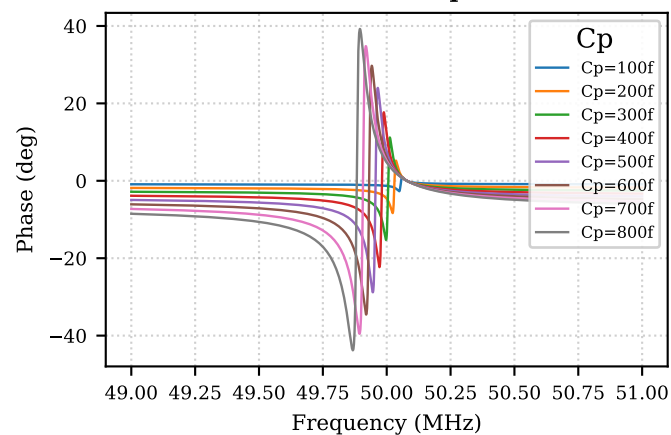
Itail = 125 μ



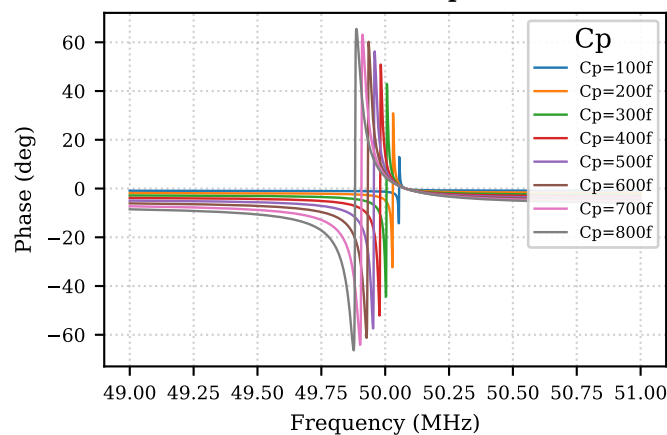
Itail = 150 μ



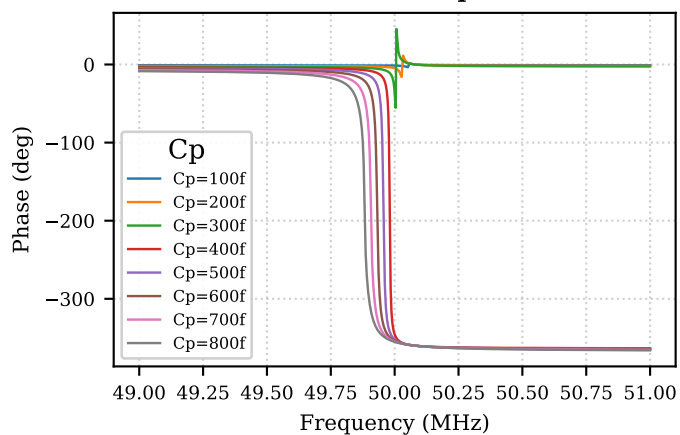
Itail = 175 μ



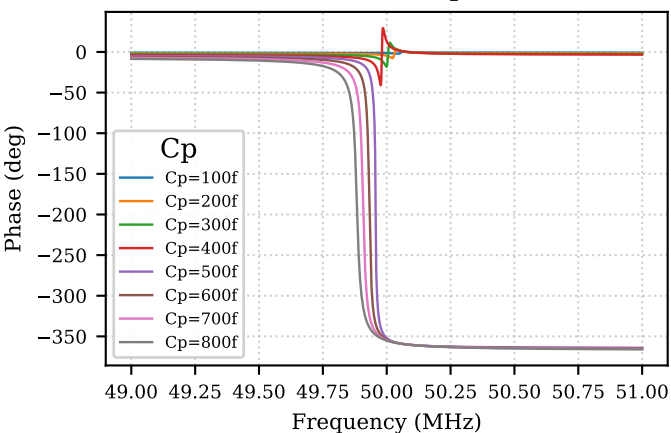
Itail = 200 μ



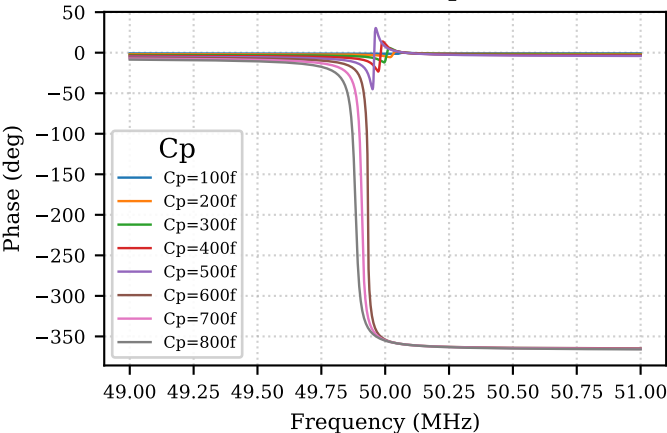
Itail = 225 μ



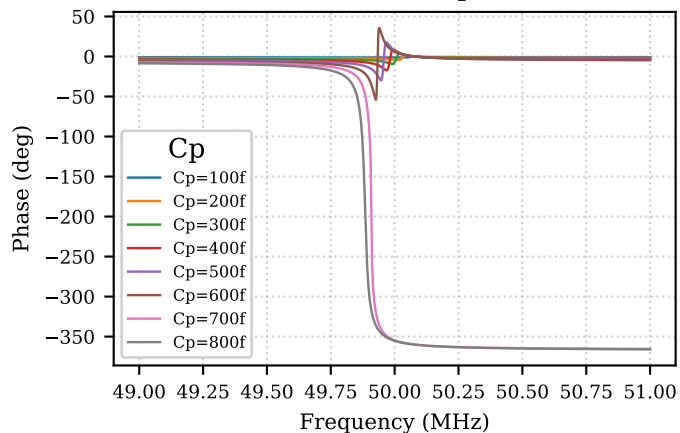
Itail = 250 μ



Itail = 275 μ



Itail = 300 μ



V(act+,act-) – Quality factor table (page 1/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	100μ	50.05	-6.022	0.1411		notch shallower than 3 d	
V(act+,act-)	100f	125μ	50.05	-6.022	0.1947		notch shallower than 3 d	
V(act+,act-)	100f	150μ	50.05	-6.022	0.2965		notch shallower than 3 d	
V(act+,act-)	100f	175μ	50.05	-6.022	0.5711		notch shallower than 3 d	
V(act+,act-)	100f	200μ	50.05	-6.022	4.135	999.6	5.007e+04	
V(act+,act-)	100f	225μ	49.68	-6.022	8.006e-05		notch shallower than 3 d	
V(act+,act-)	100f	250μ	49.38	-6.022	3.989e-05		notch shallower than 3 d	
V(act+,act-)	100f	275μ	49.18	-6.022	3.148e-05		notch shallower than 3 d	
V(act+,act-)	100f	300μ	49.03	-6.022	3.283e-05		notch shallower than 3 d	
V(act+,act-)	200f	100μ	50.03	-6.025	0.5485		notch shallower than 3 d	
V(act+,act-)	200f	125μ	50.03	-6.025	0.7496		notch shallower than 3 d	
V(act+,act-)	200f	150μ	50.03	-6.025	1.121		notch shallower than 3 d	
V(act+,act-)	200f	175μ	50.03	-6.025	2.06		notch shallower than 3 d	
V(act+,act-)	200f	200μ	50.03	-6.025	10.11	4071	1.229e+04	
V(act+,act-)	200f	225μ	49.84	-6.025	0.001264		notch shallower than 3 d	
V(act+,act-)	200f	250μ	49.61	-6.025	0.0005786		notch shallower than 3 d	
V(act+,act-)	200f	275μ	49.42	-6.025	0.0004188		notch shallower than 3 d	
V(act+,act-)	200f	300μ	49.27	-6.025	0.00036		notch shallower than 3 d	
V(act+,act-)	300f	100μ	50	-6.031	1.183		notch shallower than 3 d	
V(act+,act-)	300f	125μ	50	-6.031	1.594		notch shallower than 3 d	
V(act+,act-)	300f	150μ	50	-6.031	2.327		notch shallower than 3 d	
V(act+,act-)	300f	175μ	50	-6.03	4.046	8680	5761	
V(act+,act-)	300f	200μ	50	-6.03	14.7	8273	6044	
V(act+,act-)	300f	225μ	49.92	-6.03	0.008533		notch shallower than 3 d	
V(act+,act-)	300f	250μ	49.74	-6.03	0.002969		notch shallower than 3 d	
V(act+,act-)	300f	275μ	49.59	-6.03	0.001946		notch shallower than 3 d	
V(act+,act-)	300f	300μ	49.45	-6.03	0.001549		notch shallower than 3 d	
V(act+,act-)	400f	100μ	49.98	-6.039	1.993		notch shallower than 3 d	
V(act+,act-)	400f	125μ	49.98	-6.039	2.644		notch shallower than 3 d	
V(act+,act-)	400f	150μ	49.98	-6.038	3.757	1.41e+04	3545	
V(act+,act-)	400f	175μ	49.98	-6.038	6.186	1.753e+04	2851	
V(act+,act-)	400f	200μ	49.98	-6.038	18.26	1.392e+04	3590	

V(act+,act-) – Quality factor table (page 2/3)

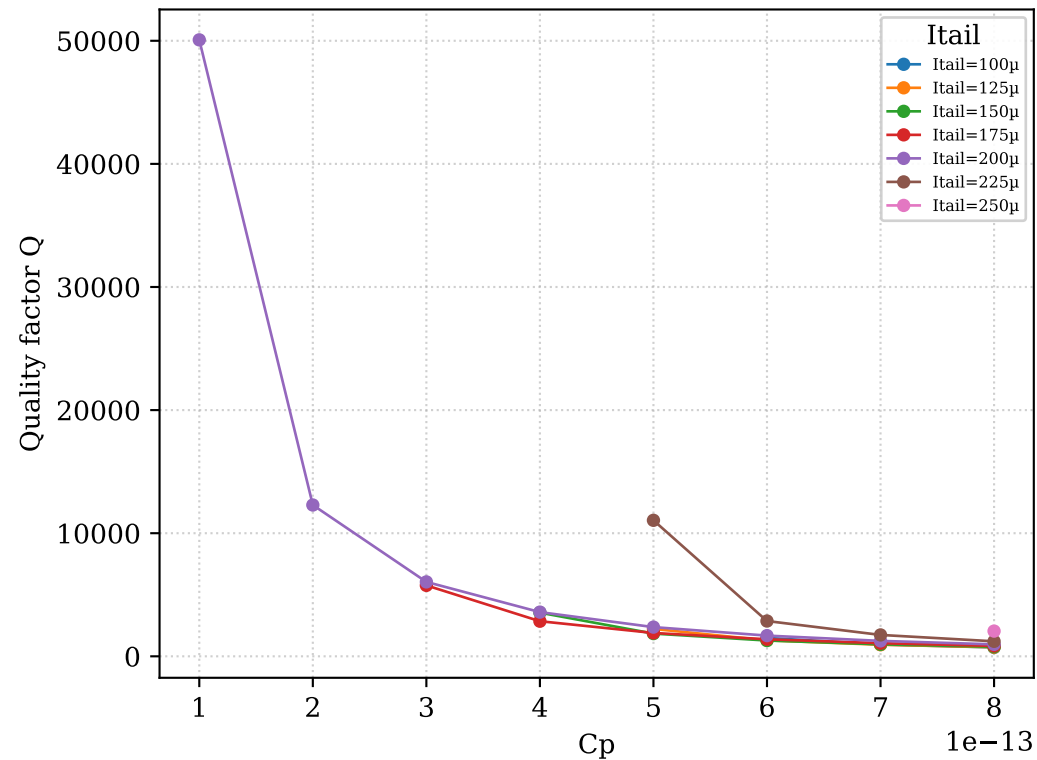
Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	400f	225μ	49.95	-6.038	0.06877		notch shallower than 3 d	
V(act+,act-)	400f	250μ	49.82	-6.037	0.01104		notch shallower than 3 d	
V(act+,act-)	400f	275μ	49.69	-6.037	0.006357		notch shallower than 3 d	
V(act+,act-)	400f	300μ	49.58	-6.037	0.004701		notch shallower than 3 d	
V(act+,act-)	500f	100μ	49.95	-6.05	2.926		notch shallower than 3 d	
V(act+,act-)	500f	125μ	49.96	-6.049	3.817	2.244e+04	2226	
V(act+,act-)	500f	150μ	49.96	-6.049	5.285	2.706e+04	1846	
V(act+,act-)	500f	175μ	49.96	-6.048	8.292	2.634e+04	1897	
V(act+,act-)	500f	200μ	49.96	-6.048	21.03	2.106e+04	2372	
V(act+,act-)	500f	225μ	49.96	-6.047	3.676	4522	1.105e+04	
V(act+,act-)	500f	250μ	49.87	-6.047	0.03885		notch shallower than 3 d	
V(act+,act-)	500f	275μ	49.77	-6.046	0.0179		notch shallower than 3 d	
V(act+,act-)	500f	300μ	49.67	-6.046	0.0121		notch shallower than 3 d	
V(act+,act-)	600f	100μ	49.93	-6.063	3.936	3.338e+04	1496	
V(act+,act-)	600f	125μ	49.93	-6.062	5.051	3.847e+04	1298	
V(act+,act-)	600f	150μ	49.93	-6.061	6.827	3.892e+04	1283	
V(act+,act-)	600f	175μ	49.93	-6.061	10.28	3.602e+04	1386	
V(act+,act-)	600f	200μ	49.93	-6.06	23.29	2.969e+04	1682	
V(act+,act-)	600f	225μ	49.93	-6.059	8.604	1.741e+04	2867	
V(act+,act-)	600f	250μ	49.9	-6.059	0.1918		notch shallower than 3 d	
V(act+,act-)	600f	275μ	49.81	-6.058	0.04926		notch shallower than 3 d	
V(act+,act-)	600f	300μ	49.73	-6.058	0.02901		notch shallower than 3 d	
V(act+,act-)	700f	100μ	49.91	-6.079	4.984	5.194e+04	960.9	
V(act+,act-)	700f	125μ	49.91	-6.078	6.3	5.303e+04	941	
V(act+,act-)	700f	150μ	49.91	-6.077	8.333	5.119e+04	975	
V(act+,act-)	700f	175μ	49.91	-6.076	12.12	4.685e+04	1065	
V(act+,act-)	700f	200μ	49.91	-6.075	25.17	3.981e+04	1254	
V(act+,act-)	700f	225μ	49.91	-6.074	12.29	2.872e+04	1738	
V(act+,act-)	700f	250μ	49.9	-6.073	2.513		notch shallower than 3 d	
V(act+,act-)	700f	275μ	49.85	-6.073	0.1572		notch shallower than 3 d	
V(act+,act-)	700f	300μ	49.78	-6.072	0.06981		notch shallower than 3 d	
V(act+,act-)	800f	100μ	49.88	-6.098	6.043	6.895e+04	723.4	

V(act+,act-) — Quality factor table (page 3/3)

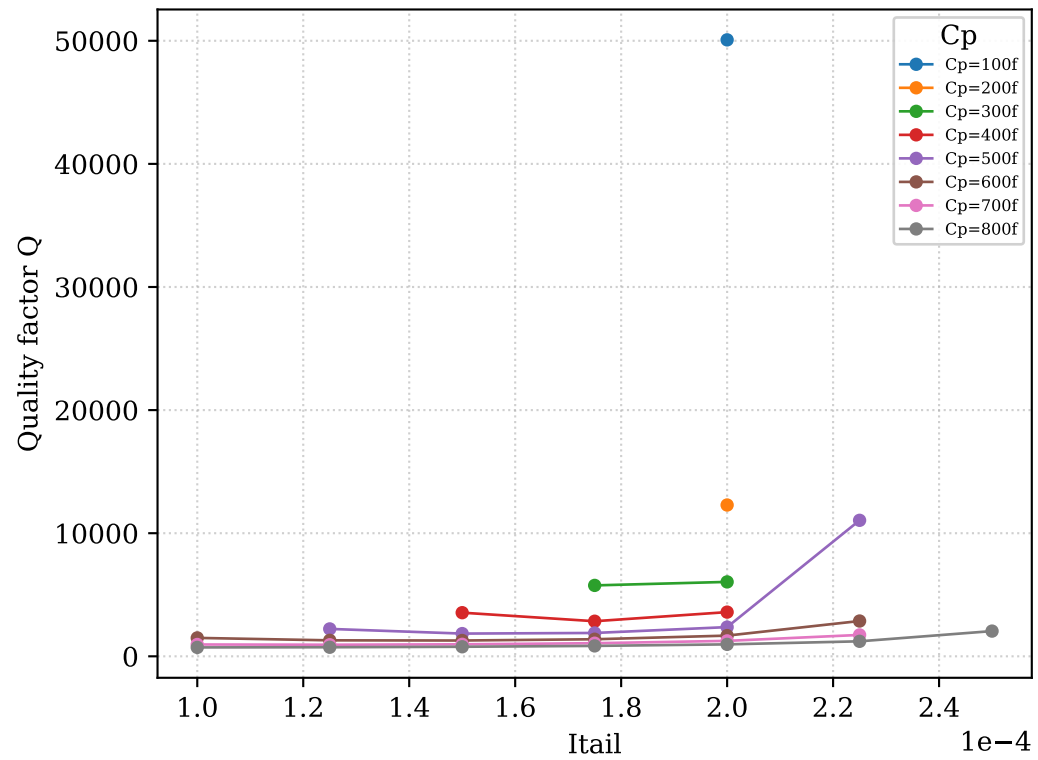
Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	800f	125μ	49.88	-6.096	7.533	6.771e+04	736.7	
V(act+,act-)	800f	150μ	49.88	-6.095	9.776	6.43e+04	775.8	
V(act+,act-)	800f	175μ	49.88	-6.094	13.81	5.893e+04	846.5	
V(act+,act-)	800f	200μ	49.88	-6.093	26.77	5.14e+04	970.4	
V(act+,act-)	800f	225μ	49.88	-6.092	15.28	4.093e+04	1219	
V(act+,act-)	800f	250μ	49.88	-6.091	6.184	2.435e+04	2048	
V(act+,act-)	800f	275μ	49.87	-6.09	0.9283			notch shallower than 3 dB
V(act+,act-)	800f	300μ	49.81	-6.089	0.1891			notch shallower than 3 dB

B. V(act+,act-) — Quality factor vs swept parameters

Q vs C_p

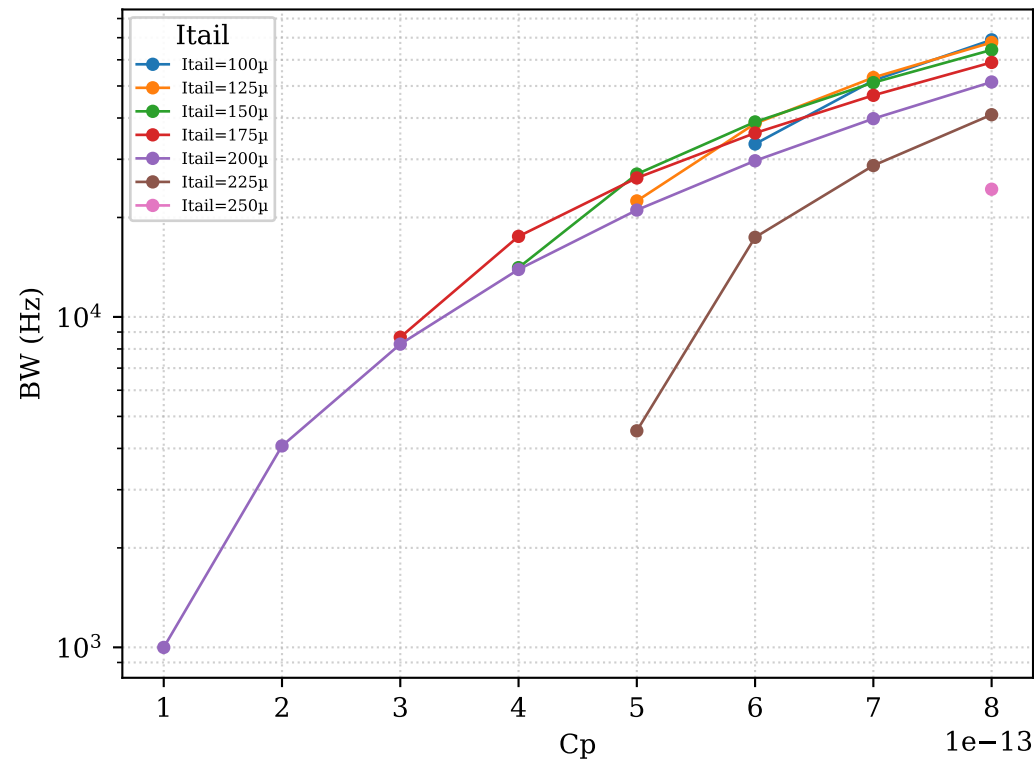


Q vs $Itail$

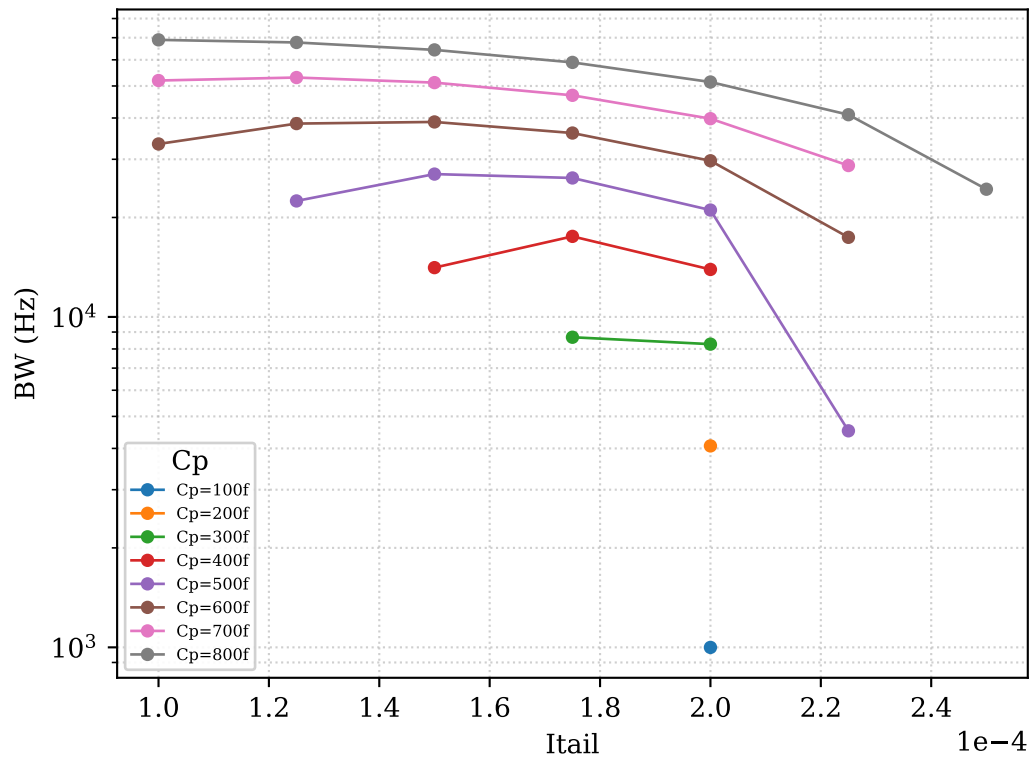


C. V(act+,act-) — Bandwidth (-3 dB) vs swept parameters

BW vs C_p



BW vs $Itail$



D. V(act+,act-) — Q heatmap (rows = Cp, columns = Itail)

	Itail=100μ	Itail=125μ	Itail=150μ	Itail=175μ	Itail=200μ	Itail=225μ	Itail=250μ	Itail=275μ	Itail=300μ
Cp=100f	no BW	no BW	no BW	no BW	50075	no BW	no BW	no BW	no BW
Cp=200f	no BW	no BW	no BW	no BW	12291	no BW	no BW	no BW	no BW
Cp=300f	no BW	no BW	no BW	5761	6044	no BW	no BW	no BW	no BW
Cp=400f	no BW	no BW	3545	2851	3590	no BW	no BW	no BW	no BW
Cp=500f	no BW	2226	1846	1897	2372	11047	no BW	no BW	no BW
Cp=600f	1496	1298	1283	1386	1682	2867	no BW	no BW	no BW
Cp=700f	961	941	975	1065	1254	1738	no BW	no BW	no BW
Cp=800f	723	737	776	847	970	1219	2048	no BW	no BW

Red = BW could not be measured | Yellow = Q computed ($Q \leq 1400$) | Green = $Q > 1400$

E. Combined quality-factor table – all signals (page 1/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	100μ	50.05	-6.022	0.1411			notch shallower than 3 d
V(act+,act-)	100f	125μ	50.05	-6.022	0.1947			notch shallower than 3 d
V(act+,act-)	100f	150μ	50.05	-6.022	0.2965			notch shallower than 3 d
V(act+,act-)	100f	175μ	50.05	-6.022	0.5711			notch shallower than 3 d
V(act+,act-)	100f	200μ	50.05	-6.022	4.135	999.6	5.007e+04	
V(act+,act-)	100f	225μ	49.68	-6.022	8.006e-05			notch shallower than 3 d
V(act+,act-)	100f	250μ	49.38	-6.022	3.989e-05			notch shallower than 3 d
V(act+,act-)	100f	275μ	49.18	-6.022	3.148e-05			notch shallower than 3 d
V(act+,act-)	100f	300μ	49.03	-6.022	3.283e-05			notch shallower than 3 d
V(act+,act-)	200f	100μ	50.03	-6.025	0.5485			notch shallower than 3 d
V(act+,act-)	200f	125μ	50.03	-6.025	0.7496			notch shallower than 3 d
V(act+,act-)	200f	150μ	50.03	-6.025	1.121			notch shallower than 3 d
V(act+,act-)	200f	175μ	50.03	-6.025	2.06			notch shallower than 3 d
V(act+,act-)	200f	200μ	50.03	-6.025	10.11	4071	1.229e+04	
V(act+,act-)	200f	225μ	49.84	-6.025	0.001264			notch shallower than 3 d
V(act+,act-)	200f	250μ	49.61	-6.025	0.0005786			notch shallower than 3 d
V(act+,act-)	200f	275μ	49.42	-6.025	0.0004188			notch shallower than 3 d
V(act+,act-)	200f	300μ	49.27	-6.025	0.00036			notch shallower than 3 d
V(act+,act-)	300f	100μ	50	-6.031	1.183			notch shallower than 3 d
V(act+,act-)	300f	125μ	50	-6.031	1.594			notch shallower than 3 d
V(act+,act-)	300f	150μ	50	-6.031	2.327			notch shallower than 3 d
V(act+,act-)	300f	175μ	50	-6.03	4.046	8680	5761	
V(act+,act-)	300f	200μ	50	-6.03	14.7	8273	6044	
V(act+,act-)	300f	225μ	49.92	-6.03	0.008533			notch shallower than 3 d
V(act+,act-)	300f	250μ	49.74	-6.03	0.002969			notch shallower than 3 d
V(act+,act-)	300f	275μ	49.59	-6.03	0.001946			notch shallower than 3 d
V(act+,act-)	300f	300μ	49.45	-6.03	0.001549			notch shallower than 3 d
V(act+,act-)	400f	100μ	49.98	-6.039	1.993			notch shallower than 3 d
V(act+,act-)	400f	125μ	49.98	-6.039	2.644			notch shallower than 3 d
V(act+,act-)	400f	150μ	49.98	-6.038	3.757	1.41e+04	3545	
V(act+,act-)	400f	175μ	49.98	-6.038	6.186	1.753e+04	2851	
V(act+,act-)	400f	200μ	49.98	-6.038	18.26	1.392e+04	3590	

E. Combined quality-factor table – all signals (page 2/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	400f	225μ	49.95	-6.038	0.06877			notch shallower than 3 d
V(act+,act-)	400f	250μ	49.82	-6.037	0.01104			notch shallower than 3 d
V(act+,act-)	400f	275μ	49.69	-6.037	0.006357			notch shallower than 3 d
V(act+,act-)	400f	300μ	49.58	-6.037	0.004701			notch shallower than 3 d
V(act+,act-)	500f	100μ	49.95	-6.05	2.926			notch shallower than 3 d
V(act+,act-)	500f	125μ	49.96	-6.049	3.817	2.244e+04	2226	
V(act+,act-)	500f	150μ	49.96	-6.049	5.285	2.706e+04	1846	
V(act+,act-)	500f	175μ	49.96	-6.048	8.292	2.634e+04	1897	
V(act+,act-)	500f	200μ	49.96	-6.048	21.03	2.106e+04	2372	
V(act+,act-)	500f	225μ	49.96	-6.047	3.676	4522	1.105e+04	
V(act+,act-)	500f	250μ	49.87	-6.047	0.03885			notch shallower than 3 d
V(act+,act-)	500f	275μ	49.77	-6.046	0.0179			notch shallower than 3 d
V(act+,act-)	500f	300μ	49.67	-6.046	0.0121			notch shallower than 3 d
V(act+,act-)	600f	100μ	49.93	-6.063	3.936	3.338e+04	1496	
V(act+,act-)	600f	125μ	49.93	-6.062	5.051	3.847e+04	1298	
V(act+,act-)	600f	150μ	49.93	-6.061	6.827	3.892e+04	1283	
V(act+,act-)	600f	175μ	49.93	-6.061	10.28	3.602e+04	1386	
V(act+,act-)	600f	200μ	49.93	-6.06	23.29	2.969e+04	1682	
V(act+,act-)	600f	225μ	49.93	-6.059	8.604	1.741e+04	2867	
V(act+,act-)	600f	250μ	49.9	-6.059	0.1918			notch shallower than 3 d
V(act+,act-)	600f	275μ	49.81	-6.058	0.04926			notch shallower than 3 d
V(act+,act-)	600f	300μ	49.73	-6.058	0.02901			notch shallower than 3 d
V(act+,act-)	700f	100μ	49.91	-6.079	4.984	5.194e+04	960.9	
V(act+,act-)	700f	125μ	49.91	-6.078	6.3	5.303e+04	941	
V(act+,act-)	700f	150μ	49.91	-6.077	8.333	5.119e+04	975	
V(act+,act-)	700f	175μ	49.91	-6.076	12.12	4.685e+04	1065	
V(act+,act-)	700f	200μ	49.91	-6.075	25.17	3.981e+04	1254	
V(act+,act-)	700f	225μ	49.91	-6.074	12.29	2.872e+04	1738	
V(act+,act-)	700f	250μ	49.9	-6.073	2.513			notch shallower than 3 d
V(act+,act-)	700f	275μ	49.85	-6.073	0.1572			notch shallower than 3 d
V(act+,act-)	700f	300μ	49.78	-6.072	0.06981			notch shallower than 3 d
V(act+,act-)	800f	100μ	49.88	-6.098	6.043	6.895e+04	723.4	

E. Combined quality-factor table — all signals (page 3/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	800f	125μ	49.88	-6.096	7.533	6.771e+04	736.7	
V(act+,act-)	800f	150μ	49.88	-6.095	9.776	6.43e+04	775.8	
V(act+,act-)	800f	175μ	49.88	-6.094	13.81	5.893e+04	846.5	
V(act+,act-)	800f	200μ	49.88	-6.093	26.77	5.14e+04	970.4	
V(act+,act-)	800f	225μ	49.88	-6.092	15.28	4.093e+04	1219	
V(act+,act-)	800f	250μ	49.88	-6.091	6.184	2.435e+04	2048	
V(act+,act-)	800f	275μ	49.87	-6.09	0.9283			notch shallower than 3 dB
V(act+,act-)	800f	300μ	49.81	-6.089	0.1891			notch shallower than 3 dB